Introduction to Algorithms and Data Structures

week 1: Introduction

SAXION UNIVERSITY OF APPLIED SCIENCES

IAD Course Description

- **Aim**: practice problem solving in Java
- **Entry requirement**: knowledge from ”Introduction to Programming”

- **Classes**:
  - 3 sessions
  - 1.5 hours per session, so 4,5 hrs/week total
- **Self study / homework**:
  - ~10 – 12 hrs / week

- **Subjects**:
  - Problem solving
  - Arrays++
  - Recursion
  - Linked Lists, Binary Trees
  - Java Collections Framework

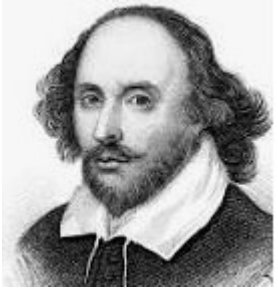
- **Test**:
  - Practical test @Saxion on Saxion systems (or BYOD), no AI, no internet (except Java API)

Learning goals

SW/ANA/1	You formulate problem solutions for simple problems in general terms.
SW/REA/1	You work with n-dimensional arrays.
SW/REA/1	You write simple recursive functions.
SW/REA/1	You use the data structures (single- and double-) linked list and (binary search-) tree in an algorithm.
SW/REA/1	You use the Map and Set data structures from the Java Collections Framework in an algorithm.
SW/ANA/1 SW/DES/1 SW/REA/1	You convert a simple problem description into a correct data structure and a corresponding algorithm.

About ChatGPT…

- *“To ChatGPT or not to ChatGPT, that is the question”* – W. Shakespeare



- *Good: "ChatGPT acting as your private tutor, aiding in your understanding and facilitating learning.“*
- *Bad: “ChatGPT does my homework for me”*

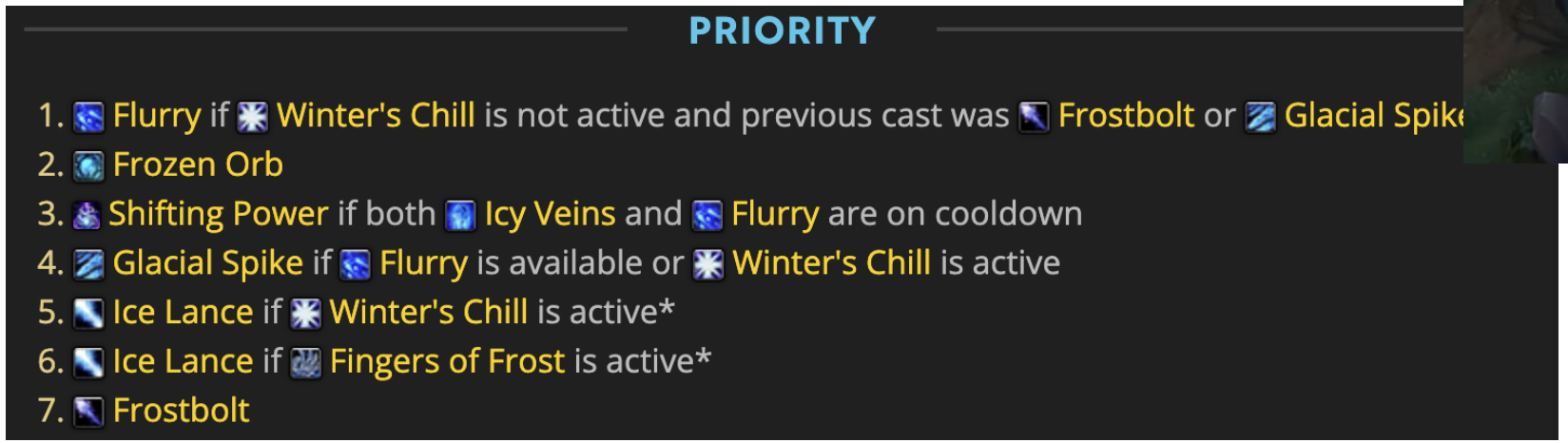
Algorithms and Data Structures

- Question: what is an **algorithm**?
  - *a set of mathematical instructions or rules that, especially if given to a computer, will help to calculate an answer to a problem (Cambridge dictionary)*
  - *a procedure for solving a mathematical problem (such as finding the greatest common divisor) in a finite number of steps that frequently involves repetition of an operation*
  - *a set of step-by-step instructions that describe how to perform a task.*
  - *a cooking recipe* ☺
- Question: Give some examples of algorithms you've used or you know.

Algorithms

- League of Legends:
  - A combo could be seen as a algorithm
    https://www.youtube.com/clip/UgkxvGYjbAWkTKgPNOT-ATY4ir2JE5tHf6bW

World of warcraft:
The order in which you should cast your spells is also an algorithm!



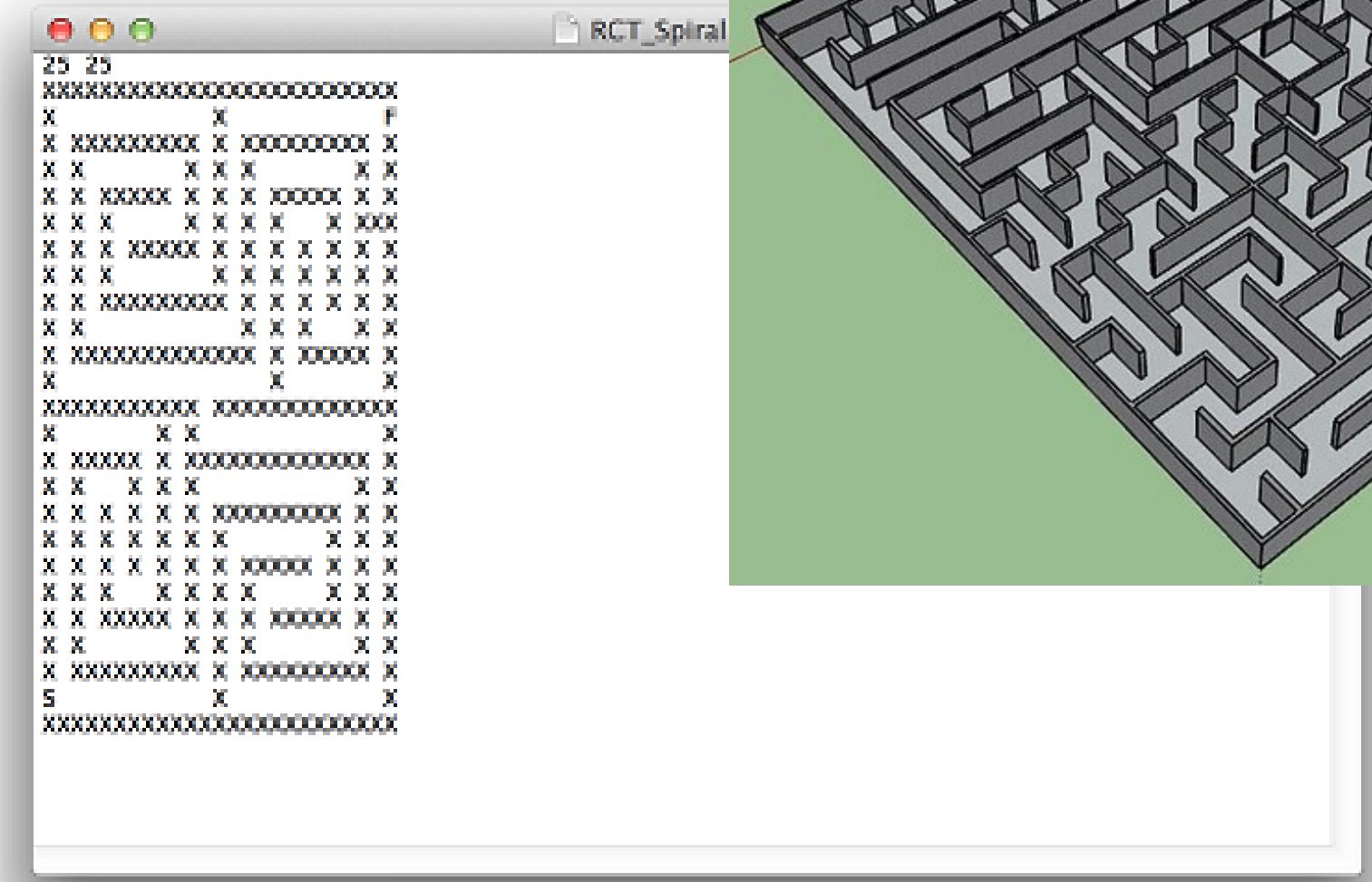
Algorithms and Data Structures

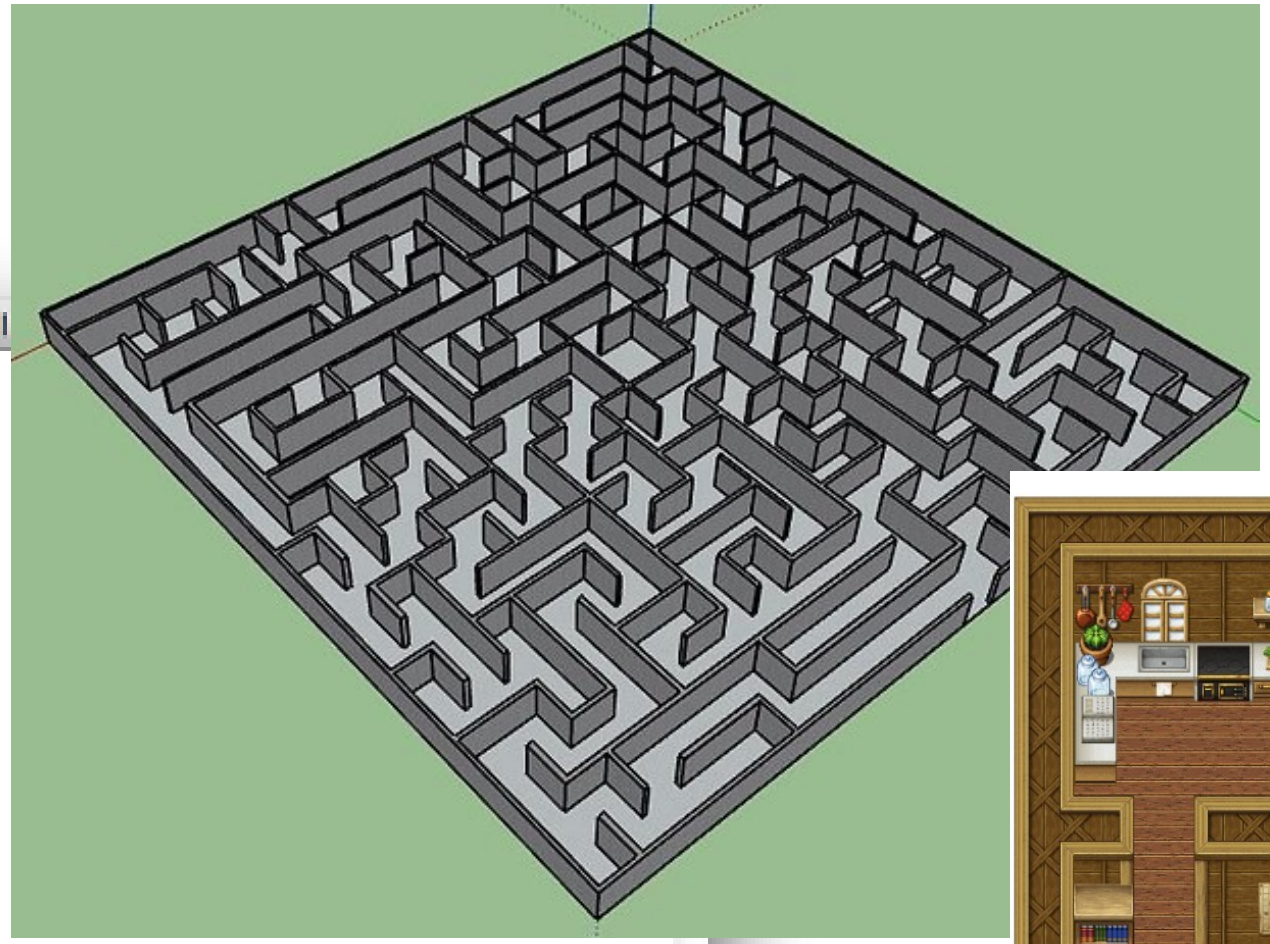
- Question: what is a **data structure**?
  - *A data structure is a storage that is used to store and organize data. It is a way of arranging data on a computer so that it can be accessed and updated efficiently. (geeksforgeeks.org)*

  - *An organization of information, usually in memory, for better algorithm efficiency (xlinux.nist.gov)*

- Question: which data structures have you used so far?
  - *e.g. variable, array*

Data Structures





Data Structures

RCT_SpiralMaze.txt

```
25 25
XXXXXXXXXXXXXXXXXXXXXXXXX
X           X           F
X XXXXXXXXX X XXXXXXXXX X
X X       X X X       X X
X X XXXXX X X X XXXXX X X
X X X     X X X X   X XXX
X X X XXXXX X X X X X X X
X X X       X X X X X X X
X X XXXXXXXXX X X X X X X
X X           X X X   X X
X XXXXXXXXXXXX X XXXXX X
X               X       X
XXXXXXXXXXX XXXXXXXXXXXXX
X       X X             X
X XXXXX X XXXXXXXXXXXXX X
X X   X X X           X X
X X X X X X XXXXXXXXX X X
X X X X X X X       X X X
X X X X X X X XXXXX X X X
X X X   X X X X     X X X
X X XXXXX X X X XXXXX X X
X X       X X X       X X
X XXXXXXXXX X XXXXXXXXX X
S           X           X
XXXXXXXXXXXXXXXXXXXXXXXXX
```

Or why not as 1 long string?

xxxxxxxxxxxxxxxxxxxxxxxxxx x Fx xxxxxxxxx x xxxxxxxxxxx xx x. x x x x x ….

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Data Structures

Or we comeup with something of our own?

‘x’25’x’1’ ‘12’x’1’ ’11’F’1’x’1’ ‘1’x’11 ….

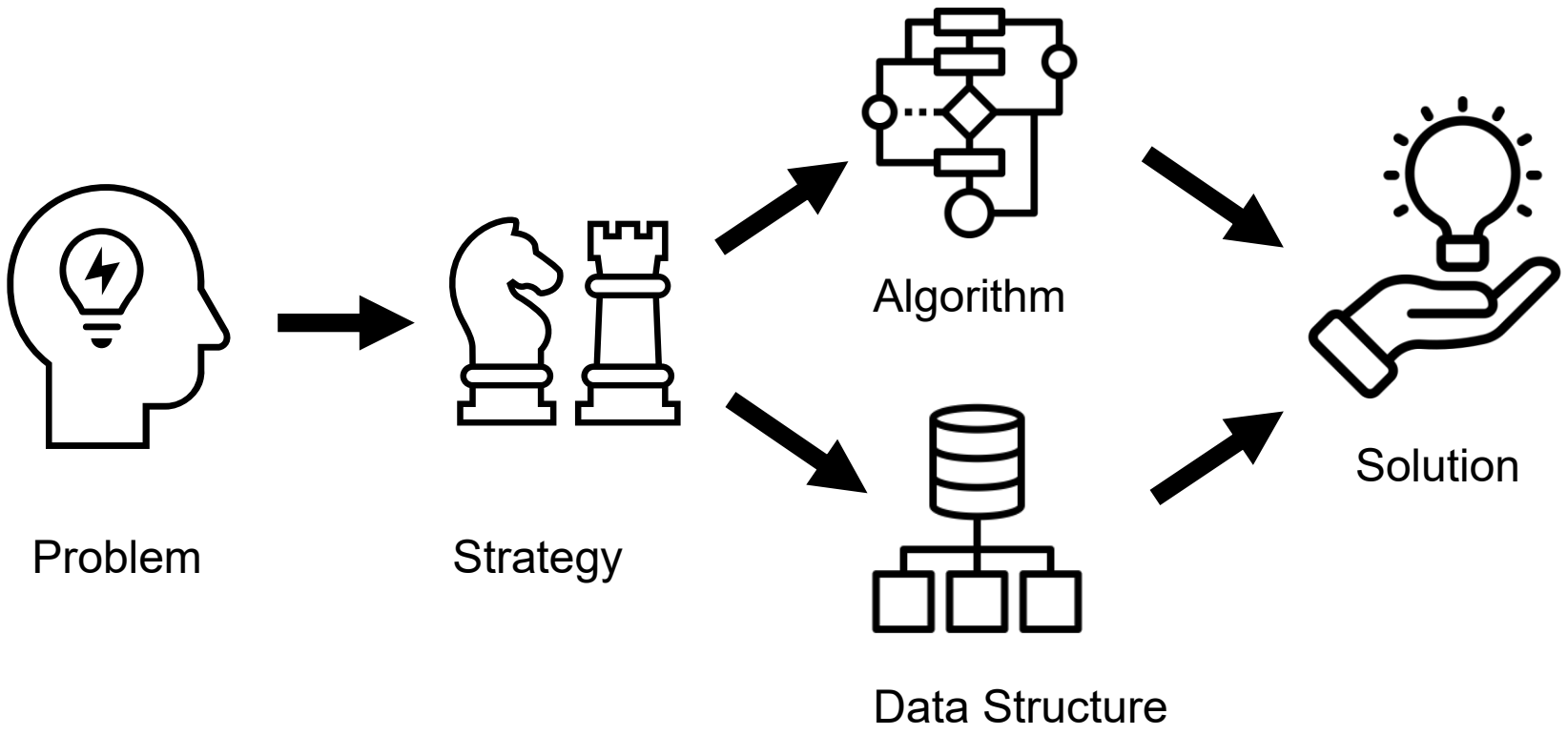
RCT_SpiralMaze.txt

```
25 25
XXXXXXXXXXXXXXXXXXXXXXXXX
X           X          F
X XXXXXXXXX X XXXXXXXXX X
X X       X X X       X X
X X XXXXX X X X XXXXX X X
X X X     X X X X   X XXX
X X X XXXXX X X X X X X X
X X X       X X X X X X X
X X XXXXXXXXX X X X X X X
X X           X X X   X X
X XXXXXXXXXXXXX X XXXXX X
X               X       X
XXXXXXXXXXX XXXXXXXXXXXXX
X        X X             X
X XXXXX X XXXXXXXXXXXXX X
X X   X X X           X X
X X X X X X XXXXXXXXX X X
X X X X X X X       X X X
X X X X X X X XXXXX X X X
X X X   X X X X     X X X
X X XXXXX X X X XXXXX X X
X X       X X X       X X
X XXXXXXXXX X XXXXXXXXX X
S           X           X
XXXXXXXXXXXXXXXXXXXXXXXXX
```

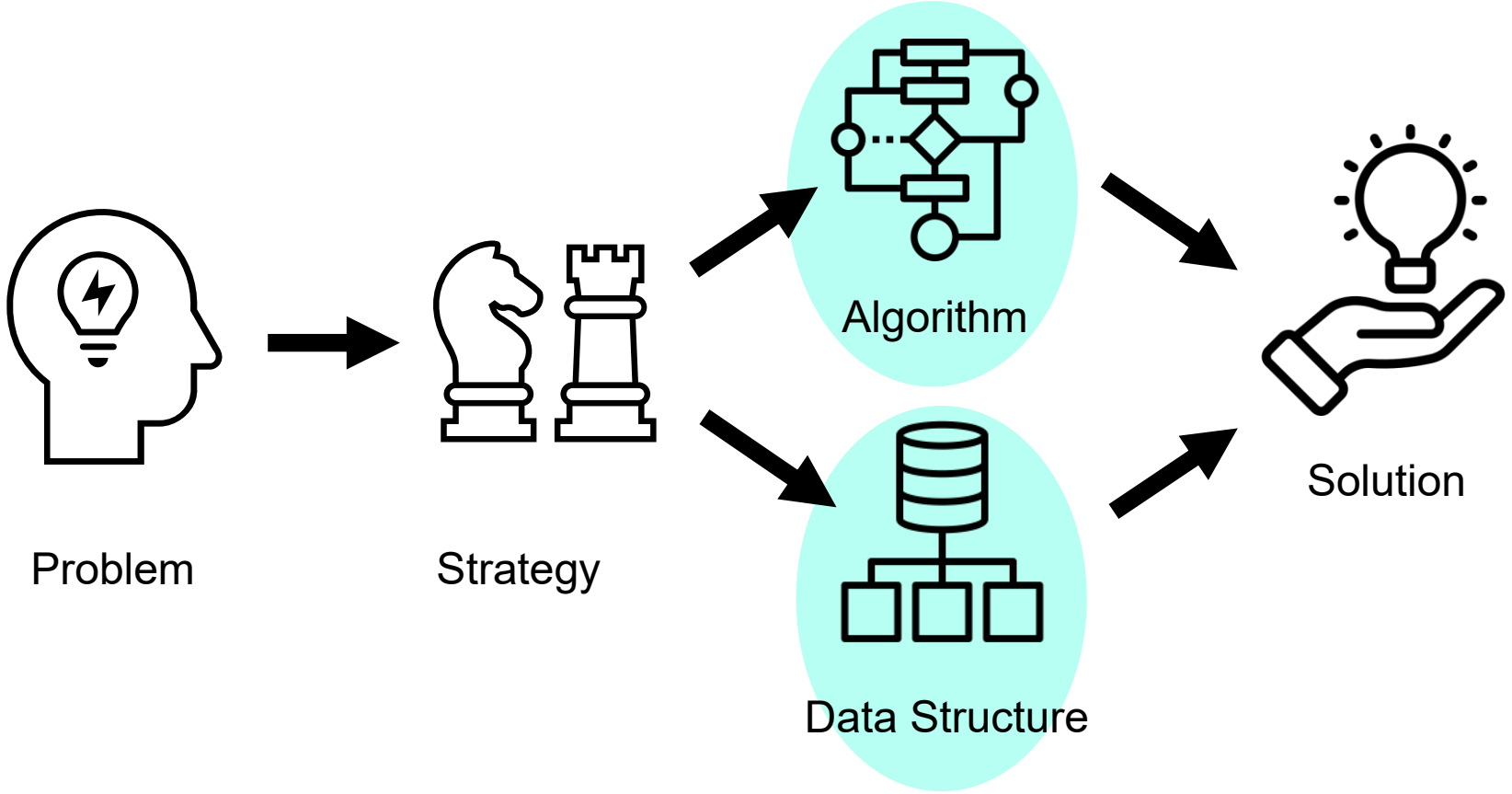
Problem Solving

- Description in *Introduction to Programming*:
  1. Breaking down a (larger) problem into smaller steps
     1. Describe the problem accurately in your own words.
     2. Identify parts you already know
     3. Write down small steps in plain English (or you own language)
     4. Reflect on your steps
  2. Translate the steps into Java code

Problem Solving – General approach



Problem Solving – Algorithm first or data structure first?



Giving precise instructions is not that easy...

Pseudo code

- Is “code” written in a kind of programming language agnostic “language” based on your own language, so a kind of broken-Java-English…
- It’s the first sketch to a solution
- Still specify fairly precisely what needs to be done, but in a language that people understand better than computers.
- So, it doesn't have to be worked out in exact detail,
  but good enough to be converted into a real programming language later.

Pseudo code of driving a car: first attempt

```
- Open the car
- Get into the car
- Start the car
- Drive away
```

Open the car?
Wait a minute…
it is locked!

Pseudo code of driving a car: improvement 1

```
- Open the car:
  - If the car is locked then
    - Press the unlock button on the remote control
- Get into the car
- Start the car
- Drive away
```

It is a bit more complicated than that

Pseudo code of driving a car: improvement 2

```
- Open the car:
  - If the car is locked then
    - Press the unlock button on the remote control
- Get into the car:
  - Open the driver's door
  - Sit down
  - Close the door
  - Fasten your seatbelt
- Start the car
- Drive away
```

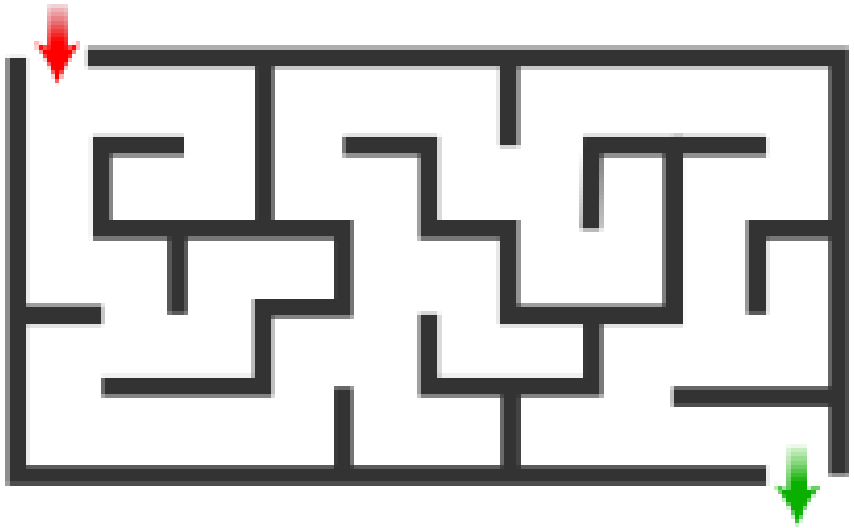
We could also explain the start a little better.

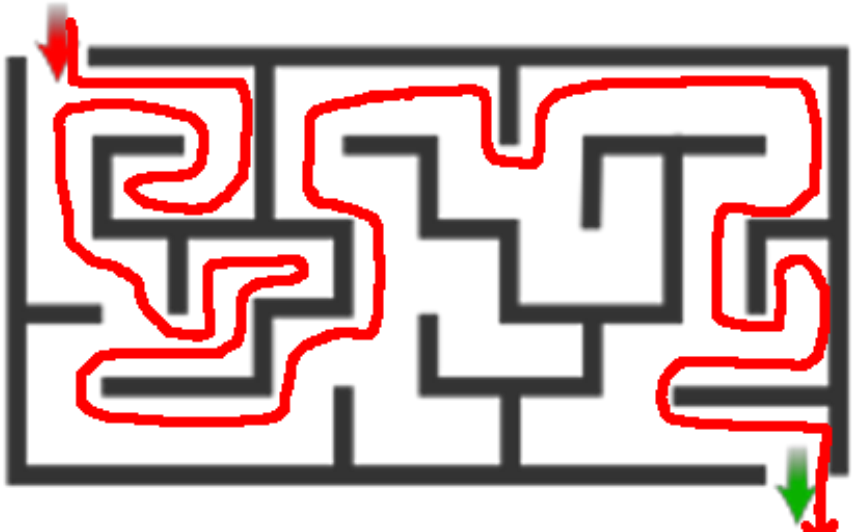
Pseudo code of driving a car: final version

```
- Open the car:
  - If the car is locked then
    - Press the unlock button on the remote control
- Get into the car:
  - Open the driver's door
  - Sit down
  - Close the door
  - Fasten your seatbelt
- Start the car:
  - Insert the key into the ignition
  - Press the clutch
  - While the engine is still off
    - Turn the key clockwise and hold for 2 seconds.
- Drive away
```

Example: Maze

- **Problem**: find the exit within a given maze, starting from the entrance
- **Strategy**: Walk through the maze; turn left when possible; keep following the left wall until the exit is reached
  - **Data structure**:
    - drawing of the maze, indicator of current position (general)
    - locations containing exits in multiple directions; indicator of current location (object-oriented)
  - **Algorithm**:
    - repeat until exit is reached:
      - if there's an opening to the left: turn left, take one step
      - else if you can take a step forward, take one step
      - else turn right





Exercise: Expert knowledge

- **Problem**: We need a system that can provide advice for a holiday destination (e.g. based on maximum travel time, means of transportation, type of housing, type of holiday, weather type, cultural interests, group size etc. etc.)
- **Strategy: ??**
  - **Data structure**?
    - *what would be a good structure to store questions / answers and decisions?*
  - **Algorithm**?
    - *how can we reach a conclusion based on preferences and stored knowledge?*

=> *Create solution with your neighbours*

Exercise: Sudoku solver

- **Problem**: complete a 9x9 grid with digits from 1 to 9, ensuring that every row, column, and 3x3 sub-grid contains each digit exactly once.
- **Strategy**: ?
  - **Data structure**?
  - **Algorithm**?

=> *Create solution with your neighbours*

5	3			7				
6			1	9	5			
	9	8					6	
8				6				3
4			8		3			1
7				2				6
	6					2	8	
			4	1	9			5
				8			7	9

Fair coin?

- **Problem**: Determine the fairness of a given coin toss

- **Strategy**?
- **Datastructure**?
- **Algorithm**?

- **Programming assignment (see Brightspace for IntelliJ project)**:
  1. Write pseudo code how to write coin toss program that stores multiple coin tosses (what algorithm and what data model?)
  2. Write a java algorithm for a coin toss
  3. Write pseudo code and THEN java code to implement an algorithm to determine the fairness of your algorithm
  4. Test: Rig the coin toss algorithm to be unfair (larger chance on heads)
  5. Re-run fairness check

=> *create the solution in duo's*

Number manipulations

Programming assignment:

How would you calculate or determine the following?

- reverse of an integer (e.g. 1234 -> 4321)
- whether an integer is a palindrome (e.g. 1331 and 23532, left-to-right the numbers are the same as right-to-left)
- whether a number is prime (only divisors are 1 and the number itself)

Write your algorithm in pseudocode, then implement it afterward.

Exercise: Converter for Roman

I	V	X	L	C	D	M
1	5	10	50	100	500	1000

- Rule: no more than 3 of the same symbols in a row
  - Allowed “subtractive” notations: IV, IX, XL, XC, CD, CM

⇒ maximum number: 3999 (MMMCMXCIX)

- Strategy?
- Data structure? Algorithm?

Exercise 1: Create algorithm (pseudocode first) to translate Arabic numbers to Roman
Exercise 2: Create algorithm (pseudocode first) to translate Roman numbers to Arabic
Exercise 3: Write java program for both

=> *create the solution in duo’s*